

Parikshit Mahajan

M.Tech (Research), Computational and Data Sciences • CGPA: 8.30

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ACADEMIC PROFILE			
M.Tech (Research)	8.30	Indian Institute of Science (IISc), Bengaluru	2027
B.Tech, Production Engineering	8.13	Veermata Jijabai Technological Institute (VJTI), Mumbai	2024
Class XII (Maharashtra Board)	76.00	Pratap College, Amalner, Jalgaon	2019
Class X (Maharashtra Board)	93.40	Sane Guruji Nutan Madhyamik Vidyalaya, Amalner	2017
COURSES			
Pattern Recognition and Neural Networks		Stochastic Models and Applications	
Advanced Deep Representation Learning		Estimation Theory	
Optimization for Machine Learning		Real Analysis	
Numerical Linear Algebra		Measure Theory and Integration	
Numerical Solution of Differential Equations		Functional Analysis	
Numerical Methods		Introduction to Scalable Systems	
PROJECTS			
AW-FNO: A Neural Operator for Parametric Partial Differential Equations (M.Tech (Res) Thesis)			Dec'25–Jul'26
- Designed and implemented AW-FNO, a hybrid deep learning model combining Fourier Neural Operator (FNO) and Wavelet Neural Operator (WNO) branches in a parallel architecture to predict solutions of parametric partial differential equations. - Developed an Adaptive Gated Fusion Mechanism, a learned gate that routes each point of the input to the better-performing branch, adding negligible parameters and remaining constant in cost as grid resolution grows. - Reduced mean relative L2 error by 45x over the FNO baseline on the Burgers benchmark and by 1.8x on Darcy flow, and achieved 57.40 dB PSNR at 4x super-resolution of turbulent flow fields. - Benchmarked the model against parameter-matched FNO and WNO baselines across four datasets using Python, PyTorch and GPU-accelerated training on an NVIDIA RTX A6000. Links: github.com/parikshit-aires-lab/AW-FNO ICCFD13 Proceedings (Paper 311)			
Retrieval-Augmented Generation: From-Scratch Implementation and Deployed Document Assistant			Aug'26–Present
- Built a Retrieval-Augmented Generation (RAG) pipeline from scratch in NumPy, implementing document chunking, embedding, vector similarity search and context assembly directly, to establish the underlying mechanics without framework abstraction. - Developing a working document assistant using LangChain and LangGraph with a Streamlit chat interface served locally, supporting document upload, semantic retrieval through an embedding model, and grounded answer generation through a chat LLM. - Comparing the two implementations on retrieval quality, latency and engineering effort, and documenting where framework abstractions add value and where they obscure failure modes.			
Unified From-Scratch Framework for Core AI/ML Architectures			Sep'26–Present
- Implementing the fundamental building blocks of machine learning and deep learning from scratch in Python and PyTorch: linear and logistic regression, ensemble methods, multi-layer perceptrons, CNNs, RNNs, Transformers, GANs and diffusion models. - Designing a single framework with a common interface, so that any architecture can be selected by name at call time and routed to its corresponding implementation, training loop and evaluation pipeline.			
PUBLICATIONS AND TALKS			
P. Mahajan*, M. Saini*, G. M. Jyrwa, D. Nagchaudhury. <i>AW-FNO: An Adaptive Approach for Fluid Flow Super-Resolution via Gated Wavelet-Fourier Learning</i>. ICCFD13, Milan, Italy, July 2026 (paper ICCFD13-2026-311). *Equal contribution; presenting author. Awarded the IISc GARP grant for conference travel. Opening Speaker, AI/ML and Trustworthy Systems Roundtable Discussion, IISc Google Day, Bengaluru, September 2026. Presented AW-FNO research on neural operators for parametric partial differential equations.			
SKILLS			
Programming: Python, C++ Tools & Libraries: PyTorch, NumPy, Pandas, scikit-learn, Matplotlib, Git, Docker, $\LaTeX$ Technical: Machine Learning, Deep Learning, Optimization, Statistical Modelling, Numerical Methods, Scientific ML			
POSITIONS OF RESPONSIBILITY			
Organizing Committee Member, CASML 2025, the International Conference on Applied AI and Scientific Machine Learning, AiREX Lab, IISc Bengaluru, December 2025. A four-day international conference held as an official pre-summit event of the AI Impact Summit 2026, with a pre-conference workshop in collaboration with Google. - Managed the conference website and led poster preparation with a three-member team. Ran the registration desk, issued delegate ID cards, and acted as an on-site point of contact for participant queries across all four days.			
ACADEMIC ACHIEVEMENTS			
GATE 2024: AIR 134 in Production and Industrial Engineering; AIR 2495 in Mechanical Engineering. - Qualified the Homi Bhabha Balvaidyanik Competition, 2016. State Rank 184, Maharashtra Olympiad Movement, 2015. - Secured ranks in the Maharashtra Talent Search Examination (MTSE) for three consecutive years, 2014–17.			
EXTRACURRICULAR ACTIVITIES			
First Prize, Pantomime, Balsahitya Sammelan, 2016. - Represented the school cricket team at district level, 2014–15, and at the district-level chess competition, 2016.			
INTERESTS AND LANGUAGES			
Interests: Cricket, chess, plays and drama. Languages: English, Hindi, Marathi.			